MULTIMODAL AI ENGINEER

Ibrohimjon Muminov

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Portfolio

M.S. in **Artificial Intelligence** researching multimodal AI, integrating **LLM**s, **VLM**s, and **computer vision** to build AI systems that **understand**, **reason**, and then **detect** and **ground** objects for more natural **human–machine interaction**.

Education

Dongguk University 09/2024 – 08/2026
Master's degree in Artificial intelligence Seoul, South Korea
GPA: 3.85/4.5
Advisor: Prof. Jihie Kim

Moscow Technological Institute 08/2016 – 06/2021
Bachelor's degree in Engineering Moscow, Russia
GPA: 4.1/5.0
Advisor: Prof. Leonid Ivanovich Munkhoev

Research Area

Foundation Models: Large Language Models (**LLMs**), **Language-Guided Object Detection**
Computer Vision: Open-Vocabulary Object Detection, **Computer Vision**
Multimodal AI: Vision-Language Models (**VLMs**), Visual Grounding, **Multimodal Reasoning**
Human-Centered AI: Human–Machine Interaction

Publications & Patents

Semantic Planning and Candidate Reasoning for Visual Grounding, 06/2026
Ibrohimjon Muminov, Jihie Kim
- **(Under Review – ACCV 2026)** Learning to Ground Like Humans: Semantic Planning and Candidate Reasoning for Visual Grounding. A cognitively inspired visual grounding framework (**See & Think, Detect, and Decide**) that models how humans resolve referring expressions through semantic planning and candidate comparison.

Vag2Det: Handling Ambiguous Prompts in Knowledge-Based Open-World Detection, 11/2025
Ibrohimjon Muminov, Jihie Kim
- **(Accepted - AICompS)** [2] *The International Conference on Artificial Intelligence Computing and Systems (AIComps 2025)*. 2025

Commonsense-Guided Open-World ObjDet Using LLMs and Visual-Semantic Matching, 05/2025
Ibrohimjon Muminov, Jihie Kim
- **(Accepted - KISP)** [1] Research and develop multimodal AI systems that leverage language reasoning to enhance open-world object detection, visual grounding, and human-centered scene understanding. The Annual Symposium of KIPS 2025 (ASK 2025). 2025

Korean Patent Application 05/2025
- *Open-World Object Detection Using a Knowledge-Based LLM and Visual-Semantic Matching*. Filed with the Korean Intellectual Property Office **(KIPO)** as Korean Patent Application No. **10-2025-0070409**, filed May 29, 2025.

Research Experience

Machine Learning Researcher, *ITRC (대학정보통신연구센터협의회)* — 11/2024 – Present, Seoul, South Korea

- Conducting research in Multimodal AI, Large Language Models (LLMs), Vision-Language Models (VLMs), visual grounding, and open-vocabulary object detection.
- Developing LLM-guided reasoning frameworks and multimodal datasets for visual grounding and language-guided object detection.

Experience

Frontend Developer (AI Applications), *iOyandasoz* — 02/2022 – 11/2022, Khujand, Tajikistan

- Develop and maintain scalable web applications that integrate AI-powered services, ensuring reliable, efficient, and user-friendly experiences.
- Integrate backend APIs for data processing and machine learning inference, building responsive interfaces to visualize model outputs and support AI-driven workflows.

Engineering Intern, *Crocus Group* — 06/2017 – 12/2017, Moscow, Russia

- Worked with 2D/3D engineering drawings and CAD models, applying geometric reasoning, spatial analysis, and visual feature interpretation relevant to computer vision applications.
- Processed and analyzed complex industrial layout models, leveraging 3D spatial understanding and geometric transformations applicable to computer vision tasks.

Selected Projects

See & Think → Detect → Decide, *Semantic Planning and Candidate Reasoning for Visual Grounding*

- **Introduced a human-like reasoning pipeline** for visual grounding through semantic planning and candidate comparison.
- **Separated grounding into four stages:** See, Think, Detect, and Decide for interpretable multimodal reasoning.
- **Integrated Qwen3-VL and GroundingDINO** using a frozen detector and lightweight LoRA adapters.
- **Enabled oracle-free visual grounding** by predicting detector classes directly from language without ground-truth object labels.
- **Achieved 85.3% Acc@0.5** on RefCOCOg-UMD with a fully explainable reasoning process.

ThinkDet, *Reducing Ambiguity in Open-Vocabulary Object Grounding*

- Analyzed failure modes of **open-vocabulary detectors under ambiguous and functional queries**.
- **Extracted semantic features from intermediate layers** of a multimodal **LLM** (InternVL).
- Integrated features into **GroundingDINO via a lightweight adapter** module with gated residual fusion.
- Achieved **+4.07pp Top-1 Hit@0.5** over the GDino baseline on a leakage-free test set.

Vague2Detect, *Handling Ambiguous Prompts in Object Detection*

- Identified limitations of **YOLO**-based models in handling vague and intent-driven queries.
- Built a semantic understanding pipeline using **SBERT** and a knowledge base for candidate expansion.
- Applied **YOLO-World** for visual verification; added an **LLM-based fallback** for unseen cases.
- Improved detection **accuracy on ambiguous** real-world scenarios over standard **OVD baselines**.

Skills

Programming: Python | C++ | JavaScript | SQL

AI & Machine Learning: LLMs | VLMs | Multimodal AI | Computer Vision | Deep Learning

Research & Development: Dataset Generation | Model Fine-Tuning | Prompt Engineering | Benchmarking & Model Evaluation

Languages

English: Fluent (Native level)

Korean: Intermediate (TOPIK Level 4)

Russian: Bilingual

Uzbek: Native

Tajik: Bilingual

Chinese: Conversational